Mask-Augmented Multimodal Animal Behaviour Recognition

Does the animal's foreground silhouette *shape* help recognise behaviour, where background segmentation maps did not?

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Abstract

Multimodal models recognise animal behaviour from video, and sometimes from audio and pose. This project asks a narrow, untested question: does adding the animal's *foreground silhouette*—its outline shape, tracked through a clip—improve behaviour recognition over video(+audio) baselines, and is any benefit concentrated in the posture- and contact-defined behaviours where shape should matter most? The motivation is a specific published result: on the MammAlps benchmark, *background/scene* segmentation maps did not help recognition, while audio gave a small lift [1]. The nearest mask-based works use segmentation only to *remove* the background or crop a region of interest [2, 3]; none treats the silhouette shape itself as a fused, positive input. The design is deliberately compute-light: every heavy encoder is frozen and its features are computed once and cached, so only a small fusion head is trained. The contribution is the answer and its analysis—per-behaviour, with inter-animal contact geometry and a background-robustness test—rather than the model wiring. A clean result is useful either way.

1 Background and motivation

Recognising what an animal is *doing* from camera footage underpins both neuroscience and ecology. Recent benchmarks combine several signals. MammAlps [1] is a multimodal, multi-view wildlife benchmark (video, audio, and reference segmentation maps; 11 activities and 19 actions across five species) with a VideoMAE-based baseline. Two of its findings frame this proposal. First, audio gives a small but real improvement over video alone (class-average mAP **0.453** → **0.473**). Second, the *reference scene segmentation maps*—a single *semantic map of the background* per fixed camera viewpoint, drawn from an animal-free photo and held essentially static across a clip—did not merely fail to help: they *degraded* recognition, dropping video from **0.453** to **0.403**, and video+audio+segmentation (0.466) likewise fell short of video+audio (0.473). Such a map encodes *where* the camera sits (habitat, terrain, vegetation): context that correlates with species and location, but says nothing about *what the animal's body is doing*; telling the model about the static background does not help it read behaviour. The distinction matters for everything that follows: MammAlps already *segments*—but it segments the *scene*, not the animal.

A parallel result comes from PanAf-FGBG [2], which pairs each chimpanzee video with a matching empty-background video and uses SAM 2 masks to *erase* the animal and synthesise backgrounds. Its purpose is to expose how strongly models lean on behaviour-correlated background, and how that hurts out-of-distribution generalisation. Crucially, the mask there is a tool for *removing* the foreground; the object of study is the background.

Pose-based pipelines (e.g. ST-GCN [9] and PoseR [10]) instead encode the animal's skeleton. Pose is compact and effective, but keypoints capture body *configuration* only sparsely—

and weakly capture two-body contact, where the relevant signal is how two bodies overlap and touch. LookAgain [4] reports exactly this asymmetry: added visual context helps some contact behaviours yet not those whose geometry pose already pins down.

The gap is the obvious next question nobody has tested. If the background does not help (MammAlps) and models over-rely on it (PanAf-FGBG), what about the one representation that throws the background away entirely and keeps only the animal's shape? This is a different *object* of segmentation: where MammAlps' map is a *static, scene-level* segmentation of the background, ours is a *per-frame foreground* (instance) segmentation of the animal itself—a silhouette whose changing outline *is* posture, and whose overlap with a second animal's mask *is* contact. Treating this *foreground silhouette as a fused, positive modality*—rather than a background-removal tool or an ROI crop—is unaddressed.

2 Research questions and hypotheses

RQ1 (does it help?)
: Does fusing silhouette-shape features improve behaviour recognition over video and over video+audio baselines?

RQ2 (where?)
: Is any improvement concentrated in posture- and contact-defined behaviours (e.g. chasing, huddling, mounting), rather than spread uniformly?

RQ3 (contact geometry)
: For two-animal behaviours, do *inter-animal* shape features (mask overlap / IoU, contact-boundary length, mutual occlusion) add signal that pose keypoints capture weakly?

RQ4 (robustness)
: Is a shape-based model more robust across *unseen camera locations* (background shift) than background-dependent video features—a direct extension of PanAf-FGBG's finding?

Hypothesis. Foreground shape carries posture and contact information complementary to pose and video. Fusing it will (i) match or exceed the audio-sized gain on the aggregate metric, (ii) show its clearest per-behaviour gains on contact/posture classes, and (iii) degrade more gracefully than video-only features under background shift.

3 Proposed approach

Datasets. *MammAlps* [1] is the primary dataset: it natively carries video, audio and behaviour labels, and supplies the published baseline to beat. *MABe22* [6] (mouse-triplet subset: clean top-view video, 12-keypoint pose, and the named contact behaviours Chase, Huddle, Oral Contact, Oral–Genital Contact, but no audio) is a controlled second setting. It is not merely a rehearsal: its multi-animal contact behaviours are where RQ3's inter-animal geometry can be tested directly, which MammAlps's largely single-animal clips cannot.

Architecture. Figure 1 shows the design. The heavy models are **off-the-shelf and pre-trained**—we run them as-is and *never update their weights*—so their outputs are computed once, offline, and cached to disk (Table 1). VideoMAE for the clip and an AST/AudioMAE-style audio model over log-mel spectrograms (MammAlps only) are *encoders*: each turns a clip into a fixed vector. SAM 2 [5] is different—it is a *segmentation* model, prompted by the dataset's animal tracks/boxes and propagated across the clip, that outputs a per-frame *mask* rather than an embedding. The silhouette is reduced to cheap, interpretable shape descriptors (area, aspect ratio, solidity, Hu moments, and, for two animals, mask IoU/overlap), with a heavier frozen-DINOv2 [7] masked-crop embedding held in reserve only if the hand-crafted features underperform. The *only* trainable component is a small fusion head: a tiny temporal encoder for the shape series, a

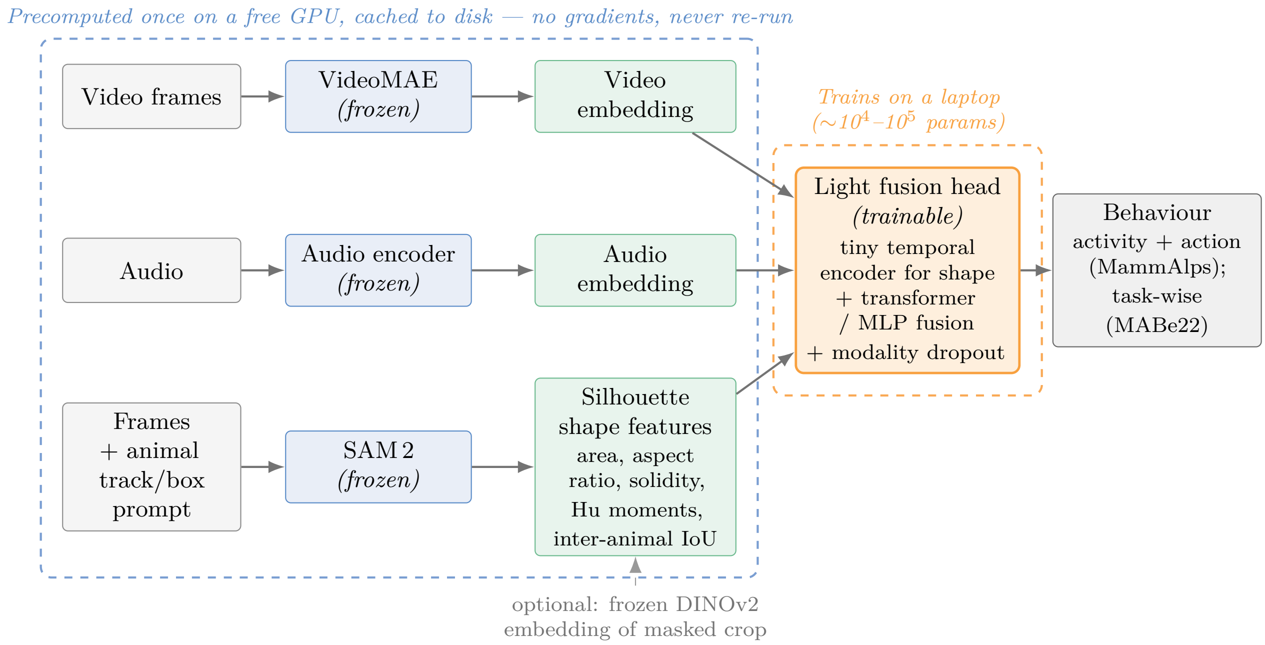

Figure 1: Initial architecture. Heavy encoders (blue) are run once and cached; only the small fusion head (orange) is trained. The mask branch defaults to cheap hand-crafted shape descriptors—roughly the footprint of pose keypoints—with a heavier DINOv2 embedding held in reserve.

one- or two-layer transformer (or even an MLP / late fusion) over the per-modality tokens, and ActionMAE-style modality dropout [8] so the mask can be dropped at test time.

Modality	Pretrained model (frozen)	Input → output
Video	VideoMAE (Kinetics)	16 frames → embedding
Audio	AST / AudioMAE-style (AudioSet)	wav → spectrogram → embedding
Shape	SAM 2 [5] + shape descriptors	frame + box → *mask* → shape numbers

Table 1: The frozen building blocks: off-the-shelf pretrained models, run once and cached, *not* trained by us. VideoMAE and AST are encoders (clip → vector); SAM 2 is a *segmentation* model whose output mask feeds the shape descriptors. Only the small fusion head (plus the tiny shape temporal encoder) is trained.

Training and protocol. Only the head trains, on cached features, on an M4 Pro (PyTorch MPS). Use official splits with a fixed seed; report *mean* ± *std over several seeds*, since on datasets this small a +0.02 mAP can be noise. Keep the head capacity identical across conditions so any gain reflects *information*, not parameters; an optional control feeds a blanked/scrambled mask (same parameters, no real shape) to confirm the silhouette is doing the work. Hierarchical heads (activity + action) on MammAlps; task-wise heads on MABe22; balanced sampling and per-class reporting throughout.

4 Novelty

The contribution is *not* a new shape representation: silhouette- and shape-based motion descriptors are long established in human action recognition (motion-energy/history templates [11]; space-time shapes [12]) and in gait recognition [13]. Our contribution is twofold: (i) we *adapt* silhouette-shape representations to *multimodal wildlife* behaviour recognition, where they are largely unexplored; and (ii) we introduce, to our knowledge, the *first use of inter-animal segmentation-mask contact geometry*—mask IoU, contact-boundary length and mutual occlusion—as explicit features for social/contact behaviour recognition. Pre-empting the obvious objection, each near neighbour is distinct:

- **MammAlps scene maps** [1]: a *background* map per viewpoint (no animal shape); shown not to help. We test *foreground* shape.
- **PanAf-FGBG** [2] (the closest work): the *same tool* (SAM 2 masks) used for the *opposite* purpose. There the mask *removes or swaps* the background, the object of study is the background itself (how much models over-rely on it), and the foreground—when kept—is still RGB *appearance*. We instead keep the mask's *geometry*, the silhouette *shape*, as a positive input, and treat their robustness finding as motivation (shape discards the very background they show models lean on). Table 2 makes the contrast explicit.
- **MAROON** [3]: mask-guided action recognition exists, but masks form background-removed cutouts (Mask-R-CNN, not SAM), with no pose and no treatment of shape as its own fused stream.
- **LookAgain** [4]: keypoint↔video fusion using raw video, no masks; its per-behaviour asymmetry motivates a dedicated shape stream for contact behaviours.

	PanAf-FGBG [2] (closest)	**This project**
Mask is used to	separate / erase / swap the *background*	derive the *silhouette shape*
Object of study	the background (bias & robustness)	the foreground shape (a feature)
Fed to the model	RGB *pixels* (FG/BG appearance)	*geometry* (outline, area, overlap)
Core question	do models over-rely on background?	does shape *add* behaviour signal?

Table 2: The closest work uses the *same* SAM 2 masks for the *opposite* purpose: it studies the background it removes, whereas we encode the foreground *shape* it would otherwise discard.

5 Expected results

The core deliverable is an ablation ladder plus a per-behaviour analysis (Table 3). The aggregate bar to beat is the audio-sized gain ($\approx +0.02$ mAP); the sharper test is per-class.

Configuration	Class-avg mAP	Reading
Video only	0.453†	baseline to reproduce
+ *their* segmentation	0.403†	background maps *degrade* video (our motivation)
+ audio	0.473†	audio adds +0.02 (the bar)
+ mask (*shape*)	**target $\gtrsim$ audio**	RQ1/RQ2: does *our* shape match/beat the audio gain?
Video + audio + mask	**?**	RQ1: does shape add *on top of* audio?

Table 3: Anticipated pattern (not measured results). †Figures reported by MammAlps [1] (Table 4); their video+audio+segmentation scored 0.466. Segmentation *lowered* video ($0.453 \to 0.403$), motivating a foreground-*shape* stream rather than a scene map. Baselines to be reproduced from cached features before any comparison is claimed.

What each outcome means.

- *Strong positive:* video+mask beats video-only by at least the audio gain, *or* a clear per-class win ($\sim +0.03$–0.05 AP) on contact/posture behaviours (chasing, mounting, Oral–Genital Contact). Headline: shape helps where background maps did not, and for the "right" (background-free) reasons—supported by the robustness test (RQ4).
- *Publishable either way:* any consistent per-class gain on contact classes, with mask-dropout robustness demonstrated.
- *Clean negative:* if shape does not help (or only where pose already suffices, echoing LookAgain), that is itself a tidy, citable result—MammAlps already showed background maps don't help;

"foreground shape doesn't either, and here is why" closes the question and directly informs whether a shape stream is worth adding to PoseR.

Pre-registered ablations. To keep a +0.02-type result interpretable against seed noise, the comparison set *and* the success criterion are fixed *before* running. We isolate: (a) **shape descriptors alone** (does shape carry any signal at all?); (b) **video vs. video+shape** (the key isolation); and (c) **full fusion**, video+audio+shape vs. video+audio (does shape add *on top of* audio?). Each condition runs over ≥ 5 seeds with class-balanced sampling and *identical* head capacity, reported as mean class-average mAP with **95 % confidence intervals**, plus per-class AP on the contact/posture classes. **Success is decided in advance:** shape helps iff (i) video+shape exceeds video with a 95 % CI excluding zero *and* a pre-registered minimum effect of +0.02 mAP (the audio-sized gain), *or* (ii) a per-class AP gain $\geq +0.03$ on contact behaviours (chasing, courtship). A blanked/scrambled-mask control (same parameters, no real shape) rules out capacity effects.

6 Feasibility

None of the heavy models is ever in the training loop: each is run once on Kaggle's free GPU and its outputs cached, so they are a bounded, one-time preprocessing cost rather than a per-step one. The trained head sees only a handful of cached vectors, so it is tiny and re-runs in minutes on a laptop—keeping the experiment in line with PoseR's lightness philosophy, whose real target is cheap *deployment* over long footage. If shape proves useful and cheap inference later matters, there are ready routes (keep the hand-crafted branch, drop the mask via the modality-dropout design, or distil), treated here as future work. The one genuinely heavy step is the single SAM pass to generate masks; this is batched, cached, and gated by a checkpoint around week 5 to reassess if MammAlps night/infrared footage makes masking too slow or noisy—at which point the headline experiment moves to the clean MABe22 setting and MammAlps becomes a stress test.

Indicative timeline

Wk 1	Read core papers; reproduce MammAlps video / video+audio baselines from cached features.
Wk 2	SAM 2 on Kaggle; generate + cache masks for MABe22 (clean case).
Wk 3	Shape-feature + inter-animal overlap extractor; sanity checks.
Wk 4	Light fusion head + modality dropout; video vs video+mask on MABe22 contact classes.
Wk 5	Cache SAM masks for MammAlps; QC mask quality; go/no-go decision.
Wk 6	Full MammAlps ablation: video $\to$ +mask $\to$ +audio $\to$ +audio+mask.
Wk 7	Missing-modality robustness; per-class & background-shift analysis.
Wk 8	Write-up; figures; package cached-feature pipeline + code.

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